28 FEB 2026

AADK TASK 1

Automated Testing Deep Dive & Mapping

MindMatrix

1. Research & Define

Unit Test

A unit test verifies a small piece of logic (usually a function or class) in isolation without involving Android framework components.
Example in Compose app: Testing whether the artwork index correctly increments when the “Next” button logic is triggered.

Instrumented UI Test

An instrumented UI test runs on an Android device/emulator and checks actual UI behavior and interactions.
Example in Compose app: Verifying that when the “Next Artwork” button is tapped, the displayed image and title update on the screen.

ComposeTestRule

ComposeTestRule is a testing utility provided by Jetpack Compose that allows us to set up composables in a test environment and interact with them (click, assert text, etc.).
Example in Compose app: Using createComposeRule() to launch the Art Space screen and simulate a button click.

Semantics Tree

The semantics tree represents the accessible structure of UI elements in Compose, which testing APIs use to find and interact with components.
Example in Compose app: Finding a button using onNodeWithText("Next") through the semantics tree to perform a click action.

2. Selected UI Feature to Test

Chosen Feature:

"Next Artwork" button in the Art Space app.
UI Elements Involved:

- Button – "Next"
- Image – Displays artwork
- Text – Displays artwork title and artist name

Expected Behavior:

When the user taps the "Next" button:

1. The artwork image should change.
2. The artwork title text should update.
3. The artist name should update.
4. The artwork index should increment internally.

3. Test Strategy Document

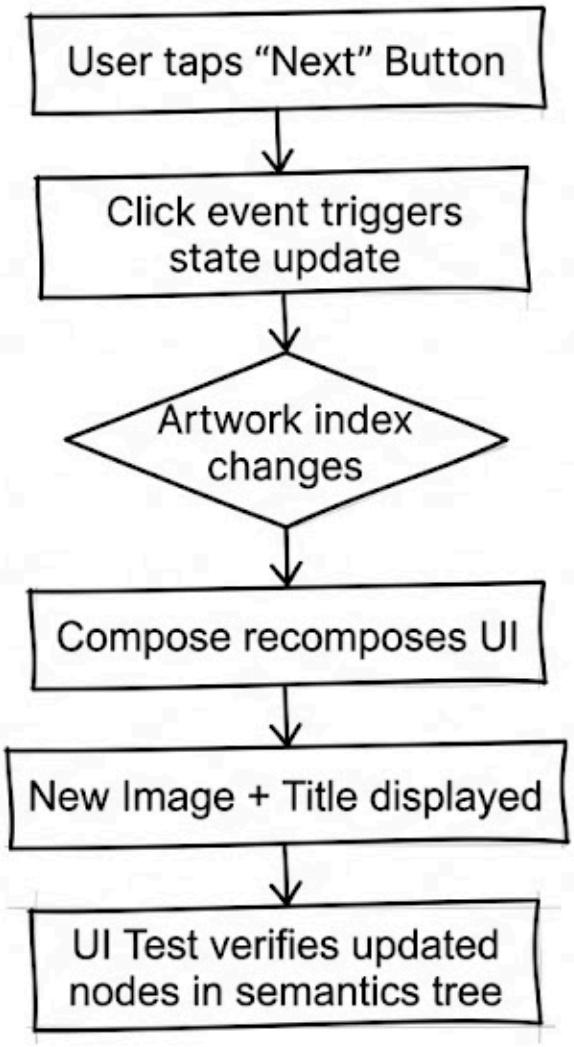
Feature: Next Artwork Button Updates UI

Test Type	What to Test	Setup Needed	Assertion / Verification
Unit Test	Artwork index increments	Call function that updates artwork	Assert new index value equals
Unit Test	Artwork list cycles back to first item	Simulate last index case	Assert index resets to 0
UI Test (Instrumented)	Button click updates title text	Launch composable using	Perform click → Assert new title is
UI Test (Instrumented)	Button click updates image	Launch composable	Perform click → Assert image
UI Test (Instrumented)	Button is clickable	Launch composable	Assert button has click action

Why Unit vs UI Test?

- Unit Test: Used for testing pure business logic like index increment and cycling because it is fast and does not need a device.
- Instrumented UI Test: Used for verifying visible UI changes after interaction because it checks real user behavior on screen.

Annotated Flow



4. Reflection

Writing tests early improves code quality because it forces us to think clearly about expected behavior before implementation. It helps catch bugs quickly and ensures features work as intended. If state changes but UI is not verified, users may see outdated or incorrect information, even if the logic is correct internally. One challenge I anticipate in writing Compose UI tests is properly identifying UI elements using semantics, especially when multiple elements have similar text or content. Understanding how recomposition works may also make testing slightly complex.